

Total No. of Questions :12]

SEAT No. :

P3246

[5670]-514

[Total No. of Pages :3

B.E. (Civil)

DAMS AND HYDRAULIC STRUCTURES
(2015 Pattern) (Semester-II) (401007) (End Sem.)

Time : 2½Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic non-programmable calculator is allowed.
- 5) Assume suitable data, if necessary.

Unit-I

Q1) State various objectives of dam safety and instrumentation. Also state instrument used to measure seepage. **[4+2]**

OR

Q2) Enlist the types of dams based on purpose. Explain any one in detail. **[3+3]**

Unit-II

Q3) What is elementary profile of gravity dam? Explain with the help of diagram, how it is modified to practical profile. **[2+2+4]**

OR

Q4) What is buttress dam? State advantages and limitations of buttress dam. **[2+3+3]**

Unit-III

Q5) What do you mean by auxiliary spillway, service spillway and emergency spillway? **[2+2+2]**

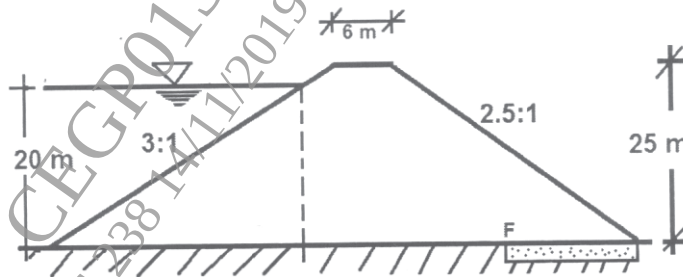
OR

Q6) What is a spillway gate? Briefly explain any two types of gates. **[6]**

P.T.O.

Unit-IV

- Q7) a)** The adjoining figure shows section of a homogeneous earthen dam. Determine the Phreatic line. Take the interval of x-coordinates as 10 m for calculations. Show the line clearly on neatly drawn section of earth dam. Take length of filter as 25m. Also find seepage discharge if $K=4 \times 10^{-6} \text{ m/s}$. **[10]**



- b)** State the corrections suggested by Khosla. Explain in detail the correction for thickness of floor for intermediate sheet pile. **[4+4]**

OR

- Q8) a)** Draw layout plan of diversion head works and label all its components and write functions of each components. **[8]**
- b)** What are different seepage control measures taken for the earthen dam for the seepage through embankment and through foundation. **[10]**

Unit-V

- Q9) a)** What are the guidelines for design of a canal alignment to optimize cost. **[8]**
- b)** Design an irrigation canal in alluvial soil according to Lacey's theory **[8]**
- i) Full supply discharge = $15 \text{ m}^3/\text{s}$
 - ii) Lacey's silt factor = 1
 - iii) Channel side slope = $\frac{1}{2}H:1V$

OR

- Q10) a)** Differentiate between design of canal by Kennedy's theory and Lacey's regime theory. **[8]**
- b)** What is canal Fall? State different types of canal falls with neat sketches. Explain any one. **[8]**

Unit-VI

- Q11)a)** Suggest a suitable cross drainage work for the following situation. Explain it with neat sketch-River bed level and high flood level is in between canal bed level and full supply level. **[2+6]**
- b)** Enlist objective of river training works. State and explain any three river training works in detail. **[2+6]**

OR

- Q12)a)** What are the cross drainage works? Explain siphon aqueduct with neat sketch. **[2+6]**
- b)** Write a short note on: **[4+4]**
- i) Guide Bunds
 - ii) Pitched island